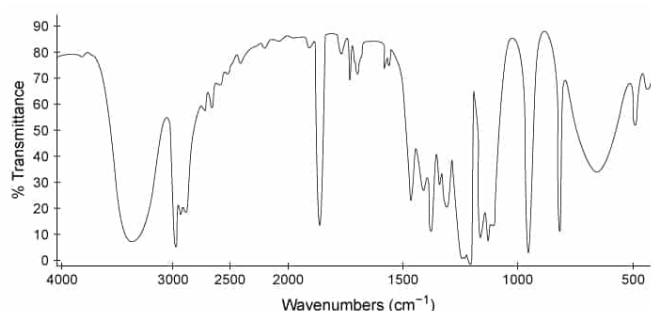


The infrared spectrum shows the frequencies of infrared light absorbed by a sample of butyraldehyde that was found to be contaminated.



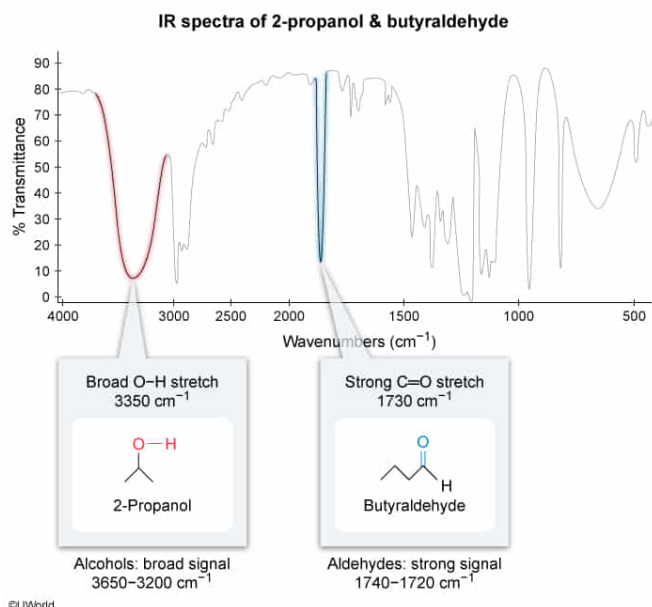
Which of the following could be the contaminant?

- ☐ A. Propene [7%]
- ☐ B. Propanone [8%]
- ☒ C. 2-Propanol [76%]
- ☐ D. Propyl ethanoate [8%]

Correct

75% answered correctly

Explanation:



In infrared (IR) spectroscopy, a sample is irradiated with IR light, and a spectrometer detects and records the percentage of radiation that passes through the sample over a range of frequencies. Different **functional groups absorb** IR radiation at **different frequencies** and **intensities**. Therefore, the signal from each group appears in a particular region of the **IR spectrum**.

Absorption frequencies depend on the type of bonds present in a particular **functional group**. Characteristic functional group absorptions include:

- 3650–3200 cm^{-1} (O–H stretch)
- 3300 cm^{-1} (sp C–H stretch)
- 3100 cm^{-1} (sp^2 C–H stretch)
- 3000 cm^{-1} (sp^3 C–H stretch)
- 1810–1650 cm^{-1} (C=O stretch)

Aldehyde carbonyls (C=O) have a characteristic strong absorption between 1740 and 1720 cm^{-1} . The sample IR spectrum shows a strong absorption at 1730 cm^{-1} , which corresponds to the carbonyl of butyraldehyde. Alcohols have a broad signal between 3650 and 3200 cm^{-1} , so the absorption at 3350 cm^{-1} most likely corresponds to an O–H stretch. 2-propanol is the only choice that contains an OH group, and therefore it is the most likely contaminant.

(Choice A) **Propene** is an alkene, which has a characteristic sp^2 C–H stretch at 3100 cm^{-1} . The sample IR spectrum does not have signals indicative of a compound with a carbon-carbon double bond.

(Choices B and D) **Propanone** contains an internal carbonyl (ketone), which has a strong C=O stretch between 1725 and 1705 cm^{-1} . **Propyl ethanoate** is an ester, and the ester C=O absorbs IR light between 1750 and 1735 cm^{-1} . The carbonyl stretch in the sample IR spectrum does not correspond to a ketone or an ester but rather to butyraldehyde.

Educational objective:

In infrared spectroscopy, a sample is irradiated with infrared light and the percentage transmission is detected and recorded over a variety of frequencies. Absorption frequencies depend on the type of bonds present in a particular functional group. Characteristic functional group absorption ranges include 3650–3200 cm^{-1} (O–H stretch), 3300 cm^{-1} (sp C–H stretch), 3100 cm^{-1} (sp^2 C–H stretch), 3000 cm^{-1} (sp^3 C–H stretch), and 1810–1650 cm^{-1} (C=O stretch).

Time spent: 0 sec

Subject:
Organic Chemistry

QID: 402650

System:
Separation Techniques, Spectroscopy, and
Analytical Methods